



INTRUSION DETECTION SYSTEM

HOW TO SETUP A HOME IDS



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CONTENTS

CONTEXT	CONCEPT	SETUP	EXAMPLE
• Why setup an IDS? • What is an IDS? • What type of IDS?	• Overall Concept; • How does it work? • IDS Requirements;	• Simple Architecture; • Installation; • Traffic Duplication; • Tuning; • Enhanced Architecture; • Going Further;	• Sample Dashboard.

WHY SETUP AN IDS?

KNOW WHEN YOUR
NETWORK/COMPUTERS ARE
UNDER ATTACK

KNOW WHEN YOUR
NETWORK/COMPUTERS HAVE
BEEN COMPROMISED

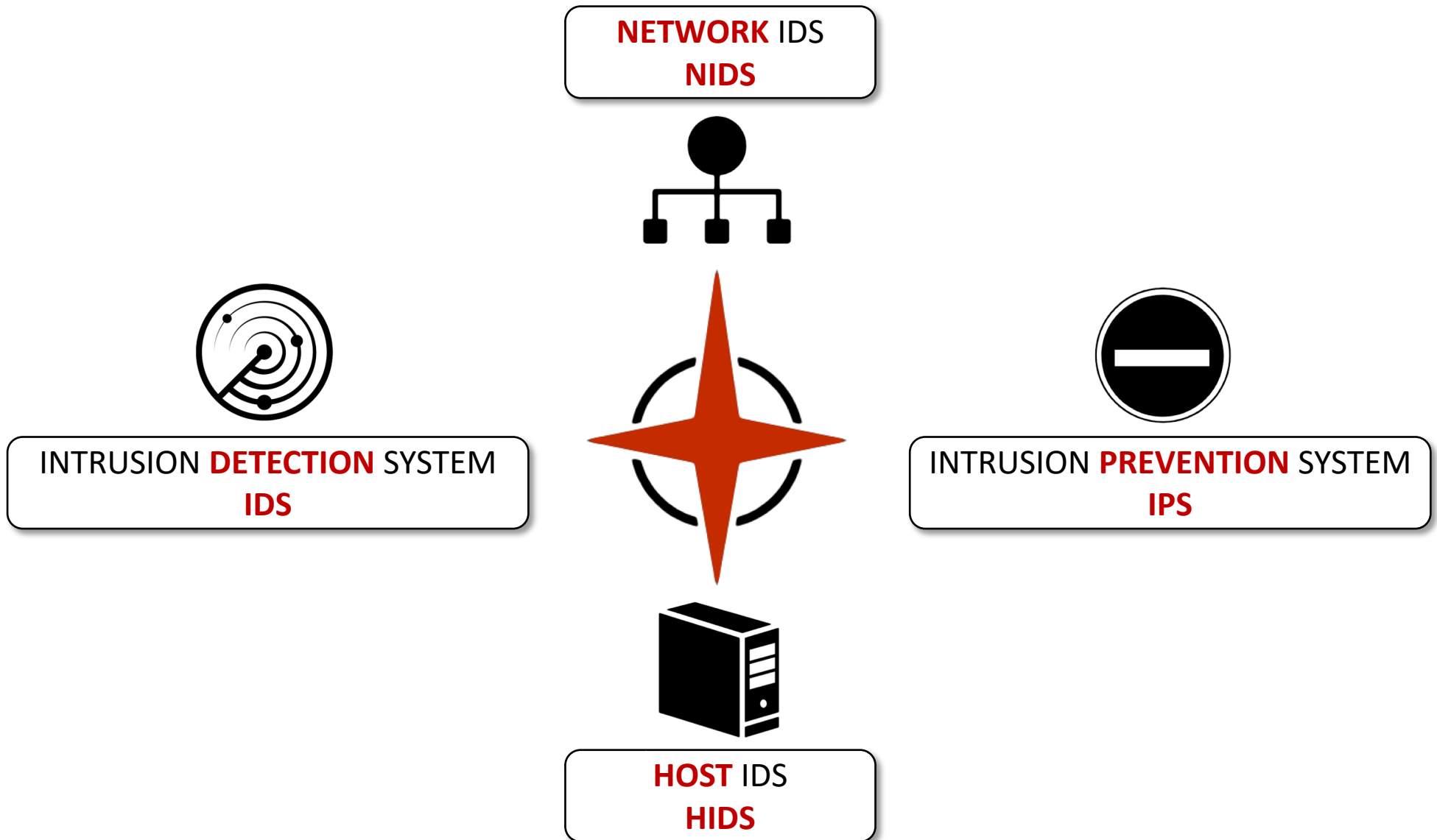
IDENTIFY SUSPICIOUS
APPLICATION ACTIVITIES

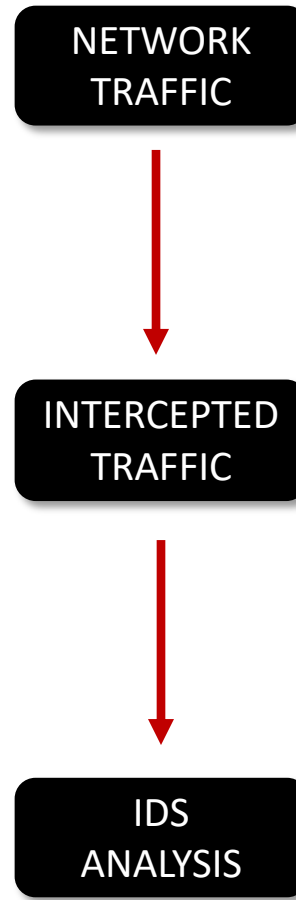
WHAT IS AN IDS?

An **Intrusion Detection System (IDS)** is a device or software application that monitors a network or systems for malicious activity or policy violations.

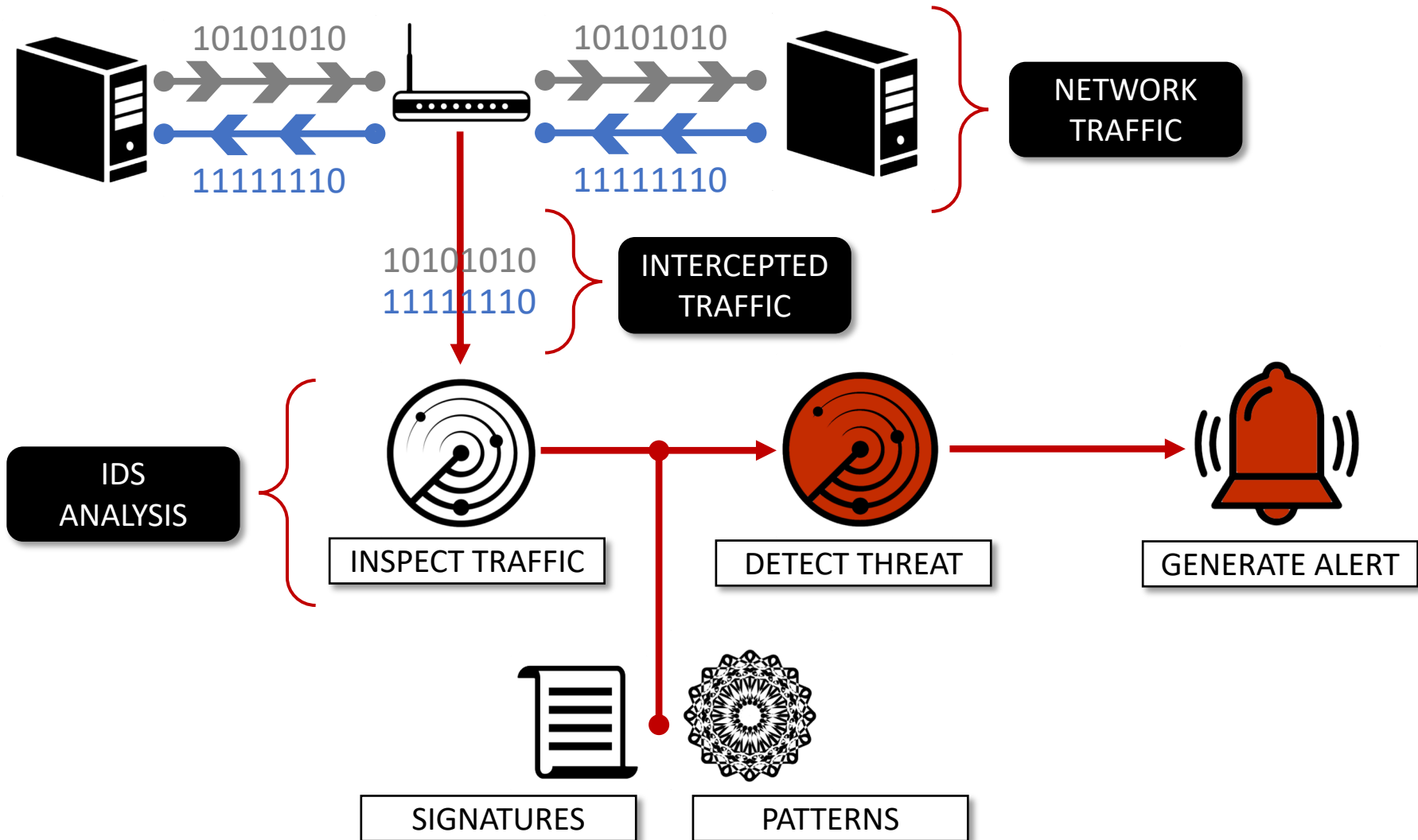


WHAT TYPE OF IDS?





HOW DOES IT WORK?



IDS REQUIREMENTS

COVERAGE

- Internal Network Activities
- (External Network -> IN)

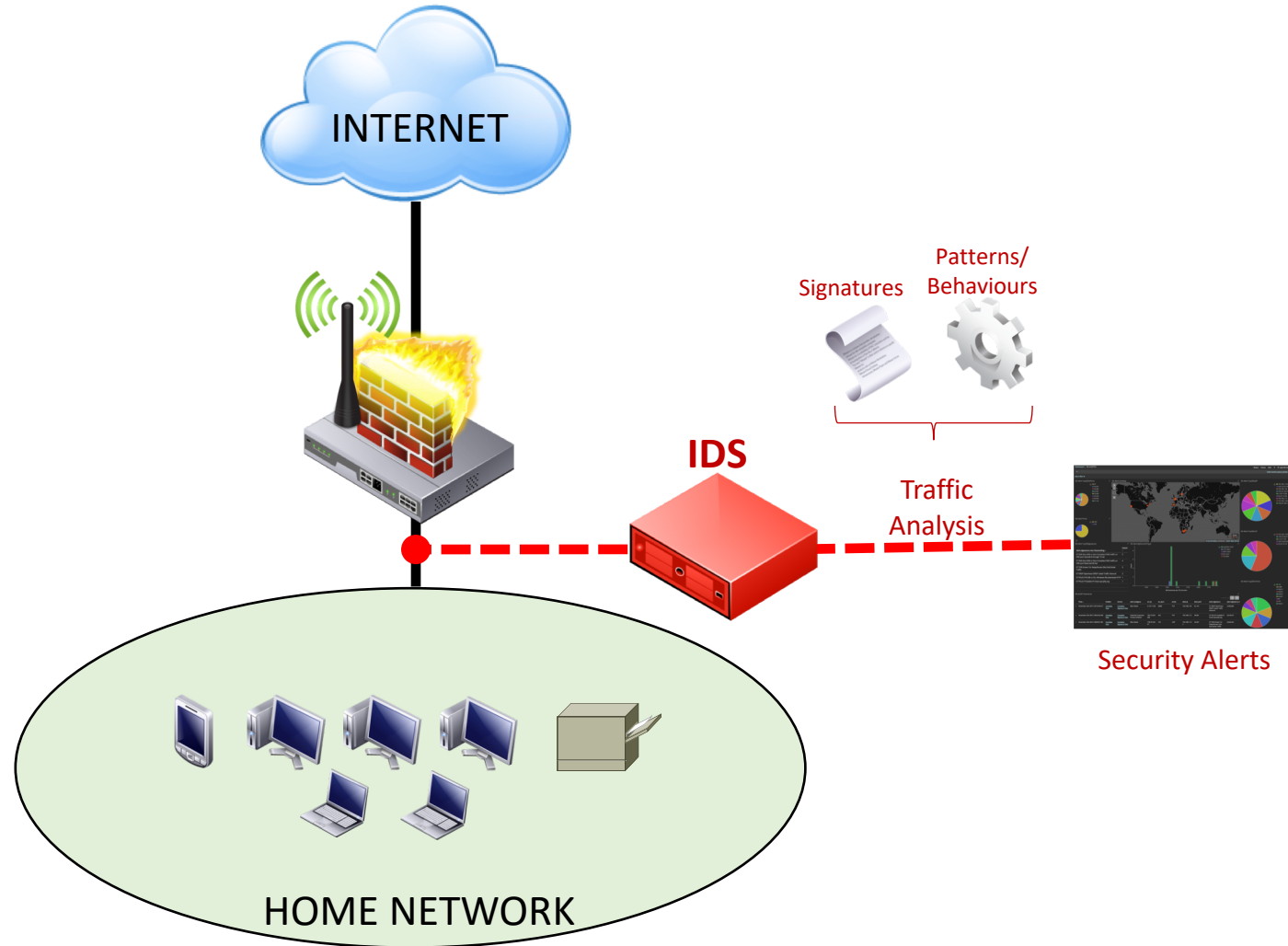
TRACKING

- What IP generated the alert?
- What Host?

TUNING

- What are you interested the most?
- Remove False Positives

SIMPLE ARCHITECTURE



1. IDS CHOICE

- Security Onion (SO)

<https://securityonion.net/>

- SELKS:

<https://www.stamus-networks.com/open-source/>

2. PLATFORM SETUP

- Virtual Machine or Dedicated Server
- 2x Network ports (Manage/Monitor)
- Enough RAM (8Gb+) and Disk Space (50Gb)
- To run 24/7

3. INSTALLATION

- Best to keep default settings
- Know in advance which network port will be your management port;
- Use testmyids.com to validate your installation

TRAFFIC DUPLICATION

1. HARDWARE

- Netgear GS105E
- Mikrotik RB2011UiAS-2HnD-IN
- Ubiquity Networks UNIFI Switch

2. DUPLICATION

- Configure a TAP, SPAN or Mirror port
- FROM: connection to your Internet Gateway
- DESTINATION: Connection to your IDS Monitoring port

2. CONSIDERATION

- Careful of VLANs -> Extra IDS configuration
- Careful of NAT -> Limits visibility

Video Link for Netgear Setup: <https://www.youtube.com/watch?v=kCSRgbEMkWs>

1. TRACKING

- NAT DISABLED
- Fixed IP or
- Reserved DHCP

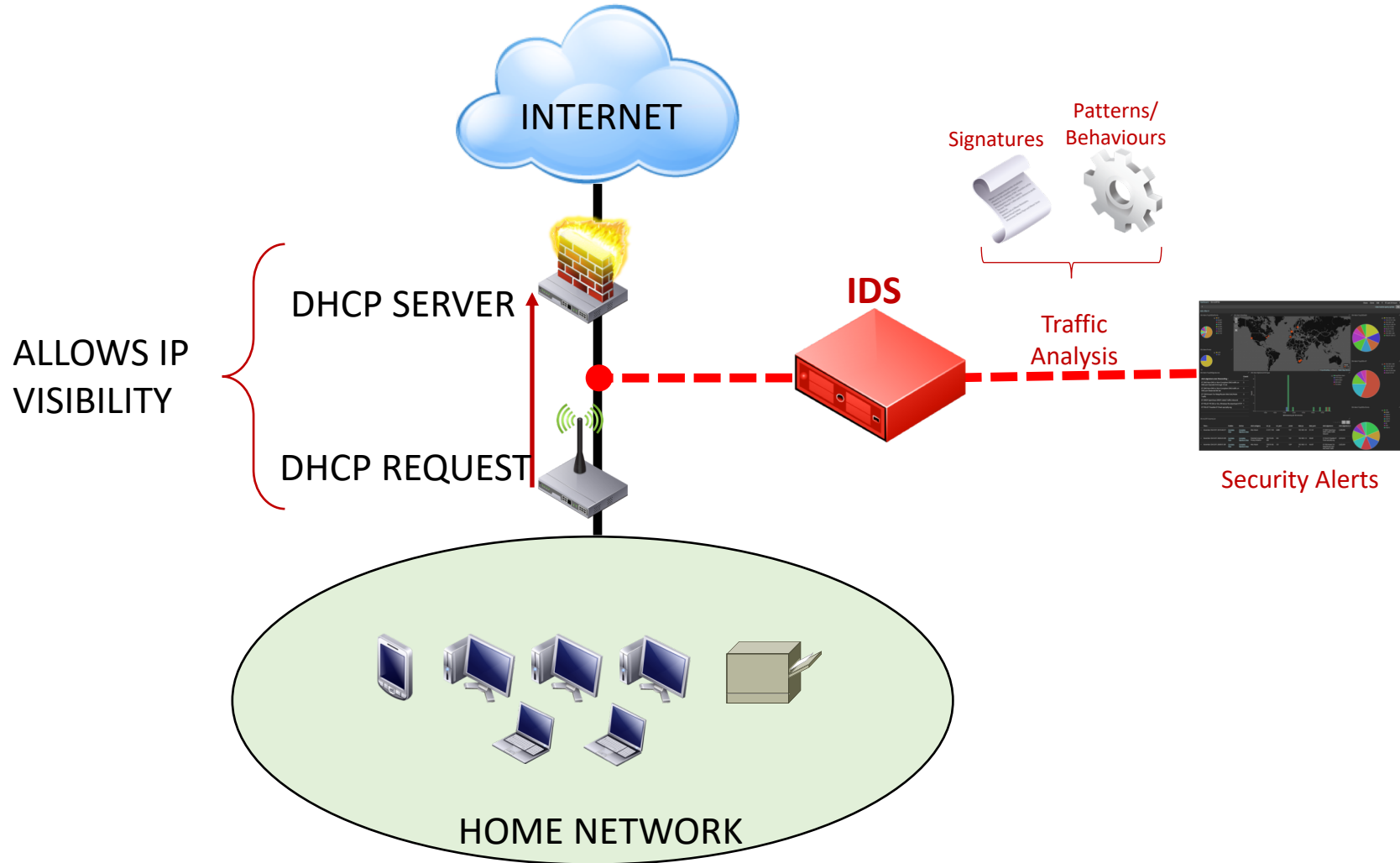
2. CONFIGURATION

- Local IP Subnet
- Define critical hosts
- Disable rules not needed

3. TUNING

- Review alerts regularly and act on false positives
- Rules Suppression
- Rules Trigger setup tuning

ENHANCED ARCHITECTURE



- **SELKS WIKI:**
<https://github.com/StamusNetworks/SELKS/wiki>
- **SECURITY ONION WIKI:**
<https://github.com/Security-Onion-Solutions/security-onion/wiki>
- **ELYSIUMSECURITY Installation Guide:**
<https://www.elysiumsecurity.com/blog/Guides/post7.html>

SAMPLE DASHBOARD

Dashboard / BUGS ALERTS

Share Clone Edit 30 seconds Last 7 days

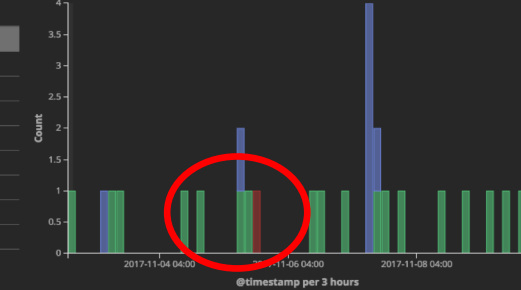
Uses lucene query syntax

Add a filter +

BUGS - Alert-Top20Signatures

alert.signature.raw: Descending	Count
ET DROP Spamhaus DROP Listed Traffic Inbound	7
ET TOR Known Tor Relay/Router (Not Exit) Node Traffic	6
ET POLICY Outdated Flash Version M1	5
GPL CHAT Jabber/Google Talk Outgoing Traffic	5
ET POLICY Possible IP Check api.ipify.org	1
ET SHELLCODE Common Windows Heap Spray String	1
ET WEB_CLIENT Obfuscated Javascript // pth	1

BUGS - Alert-ByExtraInfoType



BUGS - Alert-GeoMap



SN Alert-Count



BUGS Alert-Severity



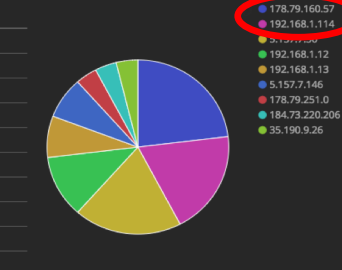
BUGS - DNS - LATEST

dns.rname.raw: Descending	Min @timestamp
tiles.maps.elastic.co	November 9th 2017, 19:42:57.147
catalogue.maps.elastic.co	November 9th 2017, 19:42:55.469
api.ipify.org	November 9th 2017, 19:42:41.350
viewposzdegzozj	November 9th 2017, 19:41:36.378
awroyist	November 9th 2017, 19:41:36.378
ndbkmp	November 9th 2017, 19:41:36.378
apis.google.com	November 9th 2017, 19:41:29.347
onedclient.sfx.ms.edgekey.net	November 9th 2017, 19:39:47.142
e9659.dspg.akamaiedge.net	November 9th 2017, 19:39:47.142
3.courier-sandbox-push-apple.com.akadns.net	November 9th 2017, 19:39:45.793

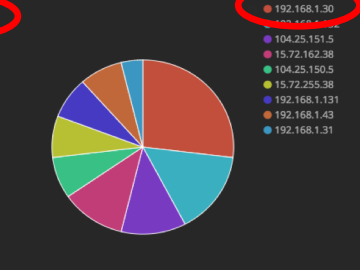
BUGS - HTTP LATEST

http.hostname.raw: Descending	Min @timestamp
ocsp2.globalsign.com	November 9th 2017, 14:17:20.910
www.airfrance.fr	November 9th 2017, 14:17:14.588
content.infos-airfrance.com	November 9th 2017, 14:17:14.010
email.airmauritiu.com	November 9th 2017, 14:13:17.582
ic.c5e23b00.088a44.gs2.sonycoment.loris-e.llnwd.net	November 9th 2017, 13:27:10.113
ic.c5e23b00.06fa54.gs2.sonycoment.loris-e.llnwd.net	November 9th 2017, 13:26:57.433
www.youmags.net	November 9th 2017, 13:23:14.170
scm.st	November 9th 2017, 13:23:12.941
gs2.wvpro.dl.playstation.net	November 9th 2017, 13:16:10.906
tmdb.np.dl.playstation.net	November 9th 2017, 13:16:09.799

SN Alert-Top20SrcIP



SN Alert-Top20DstIP



BUGS ALERT-EventsList

Time	alert.category	src_ip	dest_ip	http.hostname	alert.signature	Script	EveBox
November 9th 2017, 19:04:01	Misc Attack	5.157.7.50	192.168.1.30	-	ET DROP Spamhaus DROP Listed Traffic Inbound	Complete Signature View	Correlate Flow
November 9th 2017, 13:33:11	Misc Attack	5.157.7.50	192.168.1.30	-	ET DROP Spamhaus DROP Listed Traffic Inbound	Complete Signature View	Correlate Flow
November 9th 2017, 08:25:37	Misc Attack	5.157.7.50	192.168.1.30	-	ET DROP Spamhaus DROP Listed Traffic Inbound	Complete Signature View	Correlate Flow
November 8th 2017, 22:46:44	Misc Attack	178.79.160.57	192.168.1.132	-	ET TOR Known Tor Relay/Router (Not Exit) Node Traffic	Complete Signature View	Correlate Flow
November 8th 2017, 14:33:21	Not Suspicious Traffic	192.168.1.114	15.72.162.38	-	GPL CHAT Jabber/Google Talk Outgoing Traffic	Complete Signature View	Correlate Flow
November 7th 2017, 23:05:52	Misc Attack	5.157.7.50	192.168.1.30	-	ET DROP Spamhaus DROP Listed Traffic Inbound	Complete Signature View	Correlate Flow
November 7th 2017, 17:45:21	Misc Attack	178.79.160.57	192.168.1.132	-	ET TOR Known Tor Relay/Router (Not Exit) Node Traffic	Complete Signature View	Correlate Flow
November 7th 2017, 14:33:11	Not Suspicious Traffic	192.168.1.114	15.72.162.38	-	GPL CHAT Jabber/Google Talk Outgoing Traffic	Complete Signature View	Correlate Flow
November 7th 2017, 12:39:42	Potential Corporate Privacy Violation	192.168.1.12	104.25.150.5	www.youneedabudget.com	ET POLICY Outdated Flash Version M1	Complete Signature View	Correlate Flow
November 7th 2017, 11:46:47	Potential Corporate Privacy Violation	192.168.1.12	104.25.151.5	www.youneedabudget.com	ET POLICY Outdated Flash Version M1	Complete Signature View	Correlate Flow
November 7th 2017, 11:43:06	Potential Corporate Privacy Violation	192.168.1.13	104.25.151.5	www.youneedabudget.com	ET POLICY Outdated Flash Version M1	Complete Signature View	Correlate Flow
November 7th 2017, 11:28:09	Potential Corporate Privacy Violation	192.168.1.13	104.25.150.5	www.youneedabudget.com	ET POLICY Outdated Flash Version M1	Complete Signature View	Correlate Flow

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